

MACHINE OPERATION

AUTO MODE

- 1) Turn on the MCCB and circuit protector on the control box.
- 2) Touch panel will display AUTO MODE screen as shown in Figure 1.
- 3) Press **MASTER ON** button on the operation panel.
- 4) Then, press **ROBOT READY** switch on the touch panel to turn the robot motor power ON.
(Refer to the stated message at the message box)
- 5) Press **ROBOT ORIGIN** switch on the touch panel to origin the robot. (Refer to the stated message at the message box)

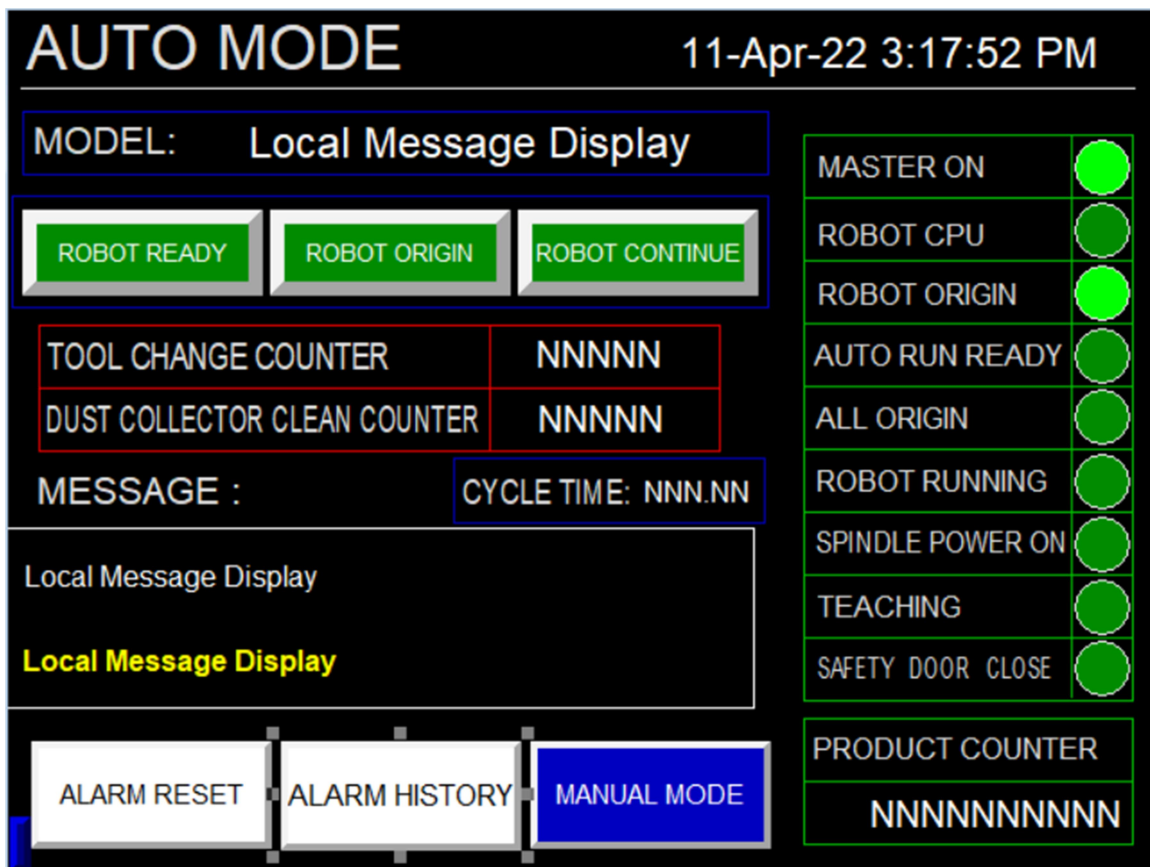


FIGURE 1: AUTO MODE SCREEN

- 6) Open the cover and place workpiece on the jig. Close the cover lock after making sure the workpiece fit on the jig.
- 7) The cover will lock after 1 second and start the process. The lock will release after inverter stop rotating.
- 8) Within the time the lock released, PL2 pilot lamp (WORKPIECE LOAD OK) will ON. This indicates that workpiece are available to be loaded or unloaded from machine.
- 9) To continue, repeat step 6 until 8.
- 10) After reset from EMERGENCY STOP during cutting, there will be an instruction at the message box to press **ROBOT CONTINUE** switch. A window screen will be pop up to ask whether to continue cut or not. Press **YES**, to continue the cutting process or **NO**, to bring the robot back to its origin position.

Note:

The message box is clearly stated the machine condition and guidance for operation step.

MANUAL MODE

- 1) Select **MANUAL MODE** on the auto mode screen.
- 2) Touch panel will display manual mode screen as shown in Figure 2. In manual mode, each of the machine's actuators (cylinders, vacuum and lamp) can be individually controlled.
- 3) Once the actuator is activated, the indicator lamp will turn ON.

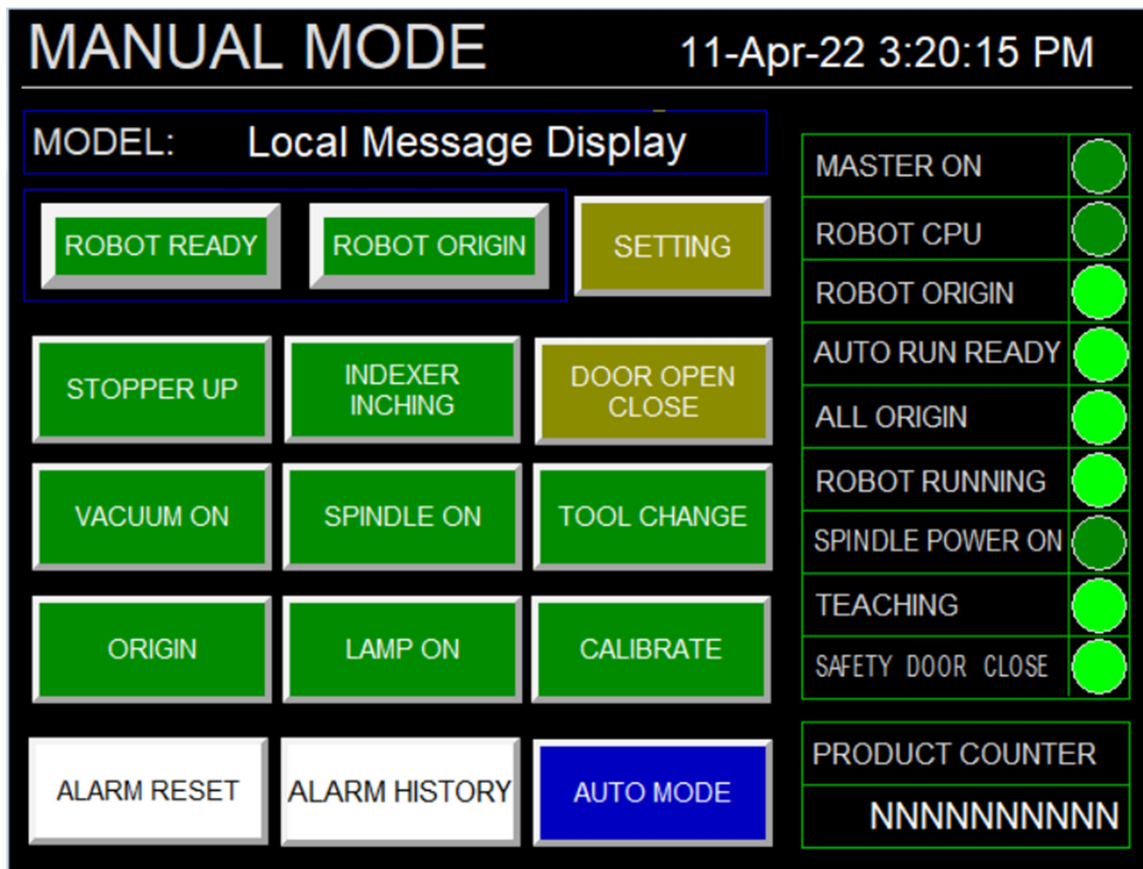


FIGURE 2: MANUAL PAGE

ORIGIN MODE

- 1) Press MASTER ON button on the operation panel.
- 2) Select the manual mode on the touch panel.
- 3) Then, press **ROBOT READY** followed by **ROBOT ORIGIN** switch before press **ALL ORIGIN** switch as shown in Figure 3.
- 4) Once the origin mode is complete, **ALL ORIGIN** switch lamp will turn **ON**.

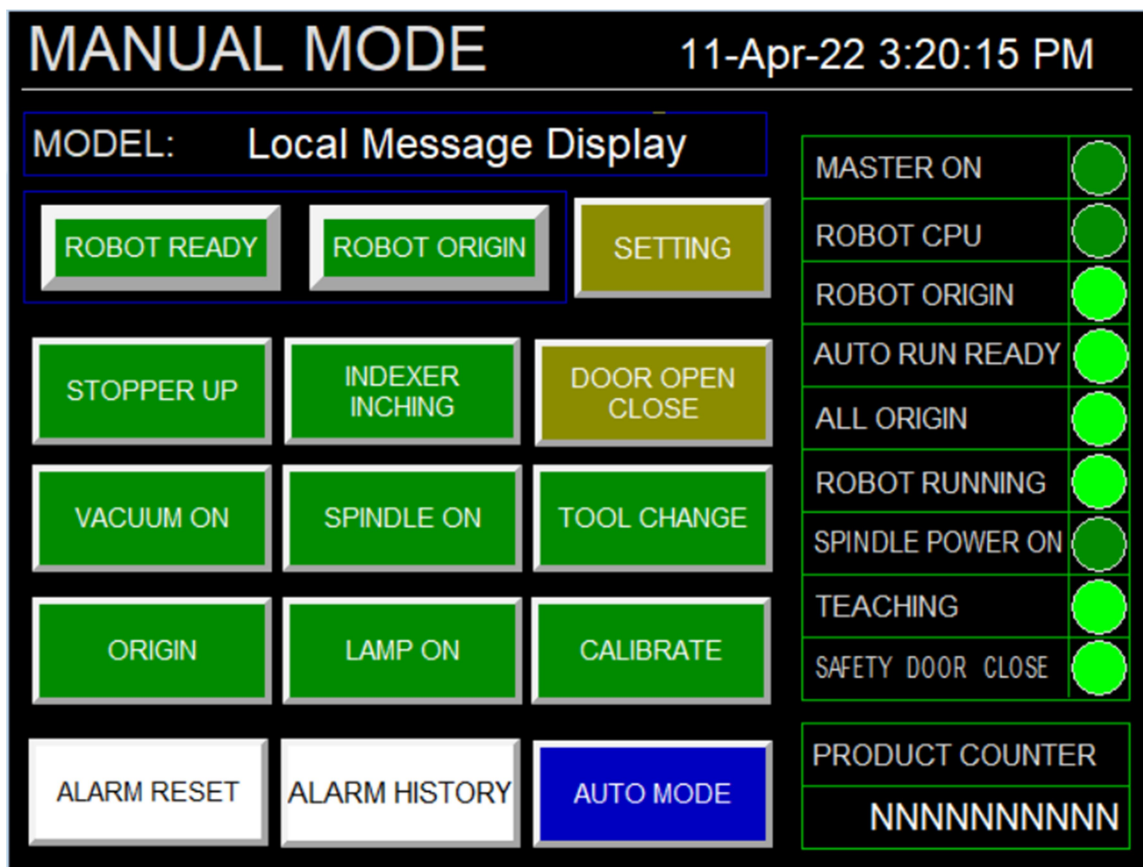


FIGURE 3: MANUAL SCREEN

SETTING

SETTING11-Apr-22 3:21:02 PM

TYPE	SETTING	CURRENT
TOTAL PRODUCT COUNTER (PCS)	NNNNN	NNNNN
TOOL CHANGE COUNTER (PCS)	NNNNN	NNNNN
DUST COLLECTOR CLEAN COUNTER (PCS)	NNNNN	NNNNN

COUNTER RESET

CHANGE SETTING

BACK

FIGURE 4: SETTING SCREEN

Figure 4 shows the setting screen. This screen displays the setting counter value and current counter value. To reset counter or change setting, press the displayed switch and password screen as shown in Figure 5 will appear.

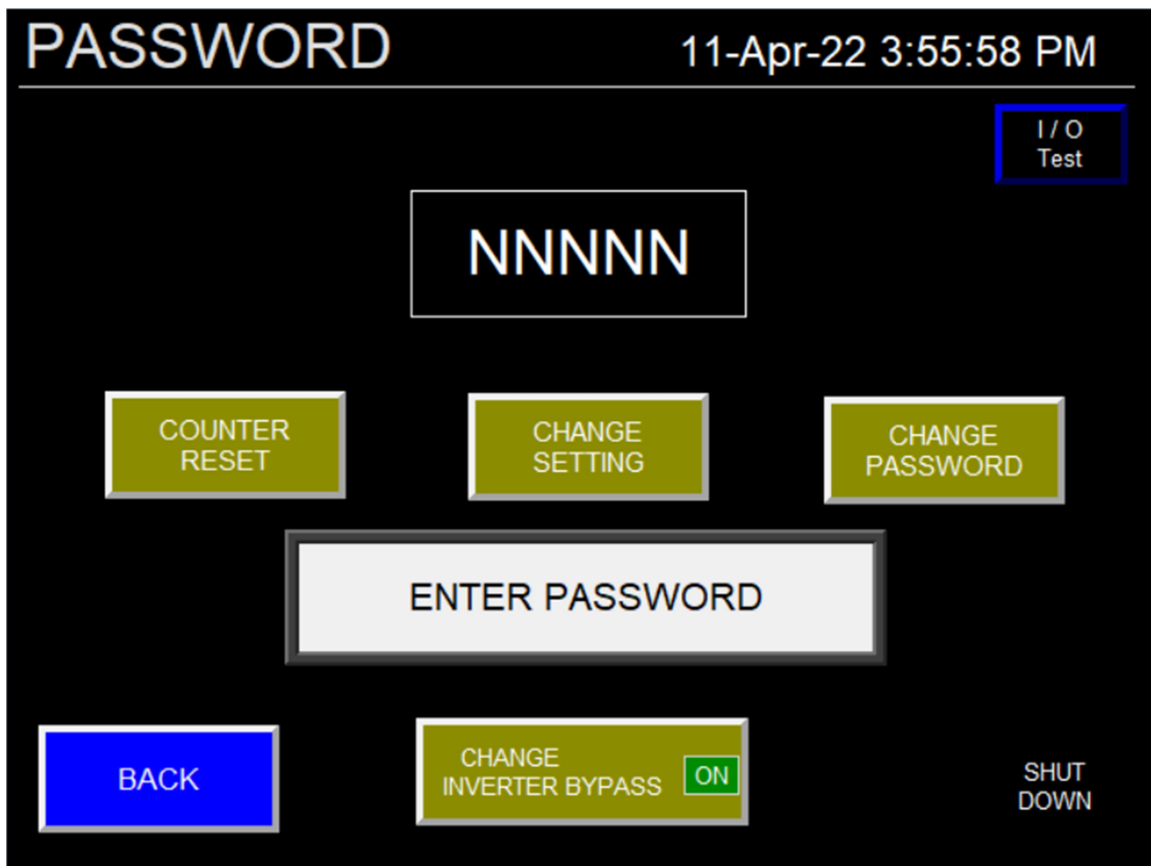


FIGURE 5: PASSWORD SCREEN

- 1) Key in the correct password to reset counter, change setting value, change password, or change inverter bypass.
- 2) Touch panel will display COUNTER RESET screen as shown in Figure 6 when COUNTER RESET switch is pressed.

COUNTER RESET

11-Apr-22 3:57:22 PM

TYPE	SETTING	CURRENT
TOTAL PRODUCT COUNTER (PCS)	NNNNN	NNNNN
TOOL CHANGE COUNTER (PCS)	NNNNN	NNNNN
DUST COLLECTOR CLEAN COUNTER (PCS)	NNNNN	NNNNN

COUNTER
RESET

ALL COUNTER
RESET

BACK

FIGURE 6: COUNTER RESET SCREEN

- 3) Press COUNTER RESET switch to clear current value achieved counter to 0.
- 4) Press ALL COUNTER RESET switch to clear all current value to 0.

ALARM HISTORY

11-Apr-22 3:57:51 PM

TIME	ALARM
* 11-Apr-22 3:57:51 PM	ABCDE FGHIJK LMNOPQ RSTUV WXYZ ABCDE FGHIJK LMNOPQ RS*

▲

▼

DELETE ALL

DELETE

ALARM RESET

BACK

Figure 7 shows the alarm history screen. Button function:

- 1) Select & delete to select the error displayed and delete.
- 2) Alarm reset to reset alarm.

ALARM

During ALARM,

- a) Buzzer will sound.
- b) Check the machine problem that occurs properly as stated on the touch panel message box or in the alarm history list.
- c) Press **RESET** switch on touch panel to reset the alarm.

Note: First press will stop the buzzer. Second press will reset the alarm.

The machine alarms will turn ON during the occurrence of conditions stated below:

No	Error contents	Cause	Countermeasure
1	Vacuum Trip	The vacuum thermal overload is tripped	1) Check vacuum thermal overload.
2	No Pressure Air	No air in system.	1) Check the air regulator. 2) Check the air supply.
3	Stopper Up Error	The cylinder cannot move up.	1) Check the air regulator. 2) Check the cylinder and valve.
4	Stopper Down Error	The cylinder cannot move down.	1) Check the air regulator. 2) Check the cylinder and valve.
5	Cover Lock Open	The cover lock is open during inverter rotating.	Check the lock that might be broken or not in a good condition. Press Alarm Reset switch on the touch panel.
6	Front/Rear Door Open	The front/rear door is open during run operation.	Close the front/rear door and press Alarm Reset switch on the touch panel.

No	Error contents	Cause	Countermeasure
7	Emergency Stop	Emergency button is pressed.	Release the emergency stop button manually and press Alarm Reset switch on the touch panel.
8	Indexer Error	Indexer adjustment is run out.	Re-do the indexer mechanical adjustment.
9	Spindle Error	Error occurs on spindle controller.	1) Check the error code on the digital speed display. 2) Call for support from Nakanishi.
10	Spindle Warning	Error occurs on spindle controller.	1) Check the warning code on the digital speed display. 2) Call for support from Nakanishi.
11	Spindle Motor Cable Unconnect	Spindle motor cable is unconnected.	Reconnect the spindle motor cable.
12	Spindle Not Auto Mode	Spindle controller is not in auto mode.	Check the PLC program.
13	Spindle Power OFF	Spindle controller is OFF.	Switch ON the spindle controller.
14	PLC Battery Low	PLC battery voltage is lower than the limit.	Replace a new PLC battery within 3 minutes which the PLC must be in power ON condition.
15	Robot Error	Robot controller alarm.	Check the error message on the robot teach pendant and refer to the robot manual.
16	Robot Battery Warning	Robot battery voltage is lower than the limit.	Switch OFF the power and replace a new robot battery. Replace both of battery one by one to prevent lose the encoder position data.

No	Error contents	Cause	Countermeasure
17	Robot Emergency Stop	Robot emergency stop button is pressed.	Release the emergency stop button manually and press Alarm Reset switch on the teach pendent and touch panel.
18	Inverter Error	Electronic thermal overload on inverter is tripped.	1) Check the inverter. 2) Press Alarm Reset switch on the touch panel.
19	Total Production Counter Achieve	Total production counter is achieved the setting limit.	Press the Counter Reset to reset it.
20	Tool Change Counter Achieve	Tool change counter is achieved the setting limit.	Press the Counter Reset to reset it.
21	Dust Collector Clean Achieve	Dust collector clean is achieved the setting limit.	Press the Counter Reset to reset it.
22	No Loading Jig	There is no loading jig on the indexer table.	Place the loading jig and press Alarm Reset switch on the touch panel.

INDICATION OF TOWER LIGHT

Colours	Details
RED	Alarm occurred. Counter achieved.
YELLOW	Machine is in MANUAL MODE or STANDBY MODE . Blinking lamp indicates warning for TOOL CHANGE .
GREEN	Machine START auto running.

TABLE 1: TOWER LIGHT